

MOHAMMAD ASHRAFUL ISLAM

Racecourse, Chittagong - 4340, Bangladesh | +8801863092843 | [Google Scholar](#) | ashrafulislam1903102@gmail.com | www.linkedin.com/in/ashraful-islam-016655200 | [Website](#).

ACADEMIC CREDENTIALS

- **Bachelor of Science in Mechanical Engineering**
- Chittagong University of Engineering & Technology (CUET), Chittagong, Bangladesh
- March 2020 - July 2025
- CGPA: 3.45/4.00 (3.84/4.00 in final year)

RESEARCH INTERESTS

Computational Fluid Dynamics (CFD) | Materials & Metallurgy | Heat Transfer | Combustion | IC Engine | Multiphase Granular Flow | Solid-Liquid Interaction | Machine Learning | Artificial Intelligence

PUBLICATIONS

Journal Articles

1. Islam, Mohammad Ashraful, Mehedi Hasan Kazol, Mizanur Rahman, Abdullah Al Rifat, and Nurul Amin. 2026. "A Comprehensive Review of Recent Advances in Biomass Pyrolysis: Feedstock Characteristics, Thermal Decomposition Mechanism, Temperature, Heating Rate, Residence Time, and Particle Size on Product Distribution". *Future Energy* 5 (2):47-72. <https://fupubco.com/fuen/article/view/912>.
2. M. A. Islam, Kazol, M. H., Rahman, M., Rahman, A., & Meraz, H. M. (2026). "Numerical investigation of biomass thermochemical conversion in a fluidized bed reactor: effect of temperature and particle size on product distribution using Eulerian-Eulerian Multiphase with KTGF". (In Peer Review – ACS Omega).
3. Haque, M. N., Al Rifat, A., Hassan, T., Alam, M. I., Islam, M. A. Islam, Kazol, M. H., Adib, A. R., & Rahman, M. (2026). "Investigation of the effects of piston bowl geometry and spray angle on combustion, performance, and emission of a diesel engine", *Energy Science & Engineering* 0 (2026): 1-19, <https://doi.org/10.1002/ese3.70614>.
4. Kazol, M. A. Islam, M. A., Rahman, M., & Rahman, M. A. (2026). "A computational investigation of varying spray angles on a dual-fuel engine with modified re-entrant piston bowl geometry". (Under Review - Heliyon).
5. A. Bakkar, M. A. Islam, and S. C. Banik, "Experimental investigation of delignified woods for sustainable piezoelectric energy harvesting," *Mater.*, vol. 13, p. 102622, Oct. 2026, <https://doi.org/10.1016/j.nxmater.2026.102622>.
6. M.A. Islam, T. Reza, Kazol, M., Rahman (2026), "Multi-Output Machine Learning and Van Krevelen Clustering for Interpretable Biomass Pyrolysis Analysis." (Under Review -next chemical)

Conference

1. M. A. Islam, M. M. Rahman, and A. Bakkar, "Fabrication and Performance Analysis of Automobile Brake Pad, using Recycled and Eco-Friendly Materials," *SCS:Engineering*, vol. 4, pp. 142–148, Aug. 2026, <https://doi.org/10.38032/scse.2026.4.67>.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant | Bio-Fuel & Combustion Lab, CUET

Sep 2025 - Present

- Numerical and experimental investigation of gas-solid reactive flows in high-temperature thermochemical conversion processes.

Undergraduate Thesis Researcher | CUET

Topic: Fabrication and performance analysis of automobile brake pads using eco-friendly and recycled materials

- Designed and tested high-temperature materials subjected to thermal and mechanical loading.
- Analyzed material degradation mechanisms under frictional heating and cyclic thermal stress.

Intern | Super Petrochemical Limited (SPL), Chittagong, Bangladesh

Duration: 3 weeks

- Observed and assisted in naphtha reforming furnaces operating at elevated temperatures, gaining exposure to industrial-scale gas-solid thermal interactions and heat transfer management.

TECHNICAL SKILLS

- CAD & Simulation: ANSYS, ANSYS Fluent, ANSYS Forte, SOLIDWORKS, AutoCAD, Aspen Plus.
- CFD Post-processing: Tecplot, ParaView, OriginPro
- Programming: Python, MATLAB, C, Fortran
- Basic Tools: LaTeX, Adobe Illustrator, Adobe Photoshop

ACHIEVEMENTS

- Merit scholarship, Department of Mechanical Engineering, CUET
- Scholarship (7th place in region), Junior School Certificate (JSC)

REFERENCES

Dr. Md. Mizanur Rahman
Professor
Department of ME
CUET, Bangladesh
Contact: +8801864800089
Email: mmrahman_me@cuet.ac.bd

Dr. Sajal Chandra Banik
Professor
Department of ME
CUET, Bangladesh
Contact: +8801821781459
Email: baniksajal@cuet.ac.bd

Dr. Abu Muhammad Sayem
Professor
Department of ME
CUET, Bangladesh
Contact: +8801710503692
Email: a.sayem@cuet.ac.bd